

Muhammad Shayan

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Profile

MSc Data Science graduate (Distinction, University of Salford) with industry experience applying NLP, LLM-based systems, and statistical modelling to real-world problems, including a GPT-4o annotation framework built in collaboration with global NGO Stop the Traffik. Equally comfortable with analyst work (SQL, Power BI, R) and ML engineering, with strong communication skills for both technical and non-technical stakeholders.

Education

University of Salford, Manchester, UK Apr 2026
MSc Data Science — Distinction

National University of Computer and Emerging Sciences (FAST), Pakistan Jun 2024
BSc Computer Science

Work Experience

Eden Smith Group — Data Science Intern (Remote) Feb 2026 – Apr 2026

- Embedded with Stop the Traffik to solve annotation quality problems in a crowdsourced English/Spanish human trafficking dataset; scoped the problem, defined metrics, and delivered a complete research pipeline.
- Conducted a baseline audit exposing critical issues: 98.2% single-annotator tasks, 26.1% average inter-annotator agreement, and 969 sub-2-second annotations.
- Designed and implemented Active-Synthesis, a GPT-4o framework using parallel sampling and multi-sample synthesis, achieving 70.7% English and 80.5% Spanish NER conflict resolution at 100% classification accuracy.
- Built an entropy-based disagreement detection pipeline with rationale extraction for human-in-the-loop review; benchmarked against Active-Prompt, LLMaAA, and Reflexion using LangGraph.
- Presented findings to NGO stakeholders, translating technical outputs into actionable recommendations.

Techverx — Product Development Intern, Pakistan Jun 2023 – Jul 2023

- Delivered competitive analysis of POS platforms (Toast, Square), benchmarking features, pricing, and UX to support roadmap prioritisation.
- Contributed to functional testing and sprint-based bug documentation across frontend, backend, and QA teams within an agile environment.

Projects

SteamRec: Recommender System at Scale PySpark, Databricks, MLlib, ALS

- Built a distributed collaborative filtering system over 200K+ user-item interactions using implicit feedback.
- Designed end-to-end ETL covering ingestion, validation, feature engineering, and model serving.
- Tuned rank, iterations, and regularisation via cross-validation and achieved sub-minute training through Spark optimisation.

Enterprise Database Design: Airport Ticketing SQL Server, T-SQL

- Designed a 3NF-normalised schema across seven modules and implemented 50+ database objects.
- Applied backup/recovery, row-level security, automated audit trails, and ETL from real-world CSV data.

Analytics Dashboard & Statistical Forecasting Power BI, DAX, R

- Built a Power BI dashboard covering 190+ countries and 15+ demographic KPIs using DAX measures and dynamic slicers.
- Forecasted AAPL stock prices using ARIMA/ETS with STL decomposition and Ljung-Box validation.
- Modelled wine quality using regression with VIF-based feature selection.

ExploreEase: Image Categorisation & Group Chat App Flutter, Flask, Firebase

- Built a full-stack mobile application using Flutter and Flask.
- Fine-tuned ResNet50 for face recognition and integrated Firebase for real-time communication.

Skills

Languages & Querying: Python, SQL (T-SQL, SQL Server), R

Machine Learning & NLP: Scikit-learn, LangChain, LangGraph, GPT-4o, Multilingual NLP, Shannon Entropy

Data & BI: Pandas, PySpark, Spark, Databricks, Power BI, DAX

Tools & Development: Git, Flask, Firebase, Flutter, ETL Pipeline Design

Professional Skills: Leadership, Team Collaboration, Communication, Adaptability, Conflict Resolution, Project Management, Time Management, Event Coordination, Decision Making